MATHS MANTHAN
• TUITION CLASSES •
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**Subject:** Mathematics | **PERIODIC TEST – 05 (POLYNOMIALS)** | **Time:** 60 Minutes
**Class:** Class X (CBSE) | **Chapter 2: Polynomials** | **MM: 30**

**Student Name:** _____ **Roll No.:** _____ **Date:** _____

**GENERAL INSTRUCTIONS:**

1. All questions are compulsory. Internal choices are provided in Section D.
2. **Section A** comprises 5 questions (4 MCQs and 1 Assertion-Reason) of 1 mark each.
3. **Section B** comprises 3 VSAQ questions of 2 marks each.
4. **Section C** comprises 3 SA questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS questions of 5 marks each.

## SECTION A: Objective & MCQs (5 × 1 = 5 Marks)

**Q1.** If $p$ and $q$ are the zeroes of the polynomial $x^2 + px + q$ where $q \neq 0$, then: [1]

(A) $p = 1, q = -2$ (B) $p = -1, q = 2$ (C) $p = 1, q = 2$ (D) $p = -2, q = 1$

**Q2.** The product of the zeroes of the polynomial $mx^2 + 6x + (2m - 1) = 0$ is $-1$. Then $m =$: [1]

(A) $\frac{1}{3}$ (B) $-\frac{1}{3}$ (C) 1 (D) $-1$

**Q3.** If $\alpha + \beta = 3$ and $\alpha^3 + \beta^3 = 7$, then $\alpha, \beta$ are the zeroes of the quadratic polynomial: [1]

(A) $9x^2 - 27x + 20$ (B) $3x^2 + 9x - 7$ (C) $2x^2 - 6x + 15$ (D) $9x^2 + 27x - 20$

**Q4.** If $\alpha, \beta$ are zeroes of $x^2 - px + q$, then the quadratic polynomial with $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ as zeroes is: [1]

(A) $qx^2 - px + 1$ (B) $qx^2 + px + 1$ (C) $px^2 - qx + 1$ (D) $x^2 - qx + p$

**Q5.** **Assertion (A):** If one zero of $4x^2 - 8kx - 9$ is negative of the other, then $k = 0$.
**Reason (R):** For negative zeroes, $\alpha + (-\alpha) = 0 \implies -\frac{b}{a} = 0 \implies b = 0$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

## SECTION B: Very Short Answer Questions (3 × 2 = 6 Marks)

**Q6.** If one zero of $(a^2 + 9)x^2 + 13x + 6a$ is reciprocal of the other, find the value of $a$. [2]

**Q7.** If $\alpha, \beta$ are the zeroes of $p(x) = x^2 + x + 1$, find the value of $\frac{1}{\alpha} + \frac{1}{\beta}$. [2]

**Q8.** If $\alpha$ and $\beta$ are the zeroes of $f(x) = 2x^2 + 5x + k$ such that $(\alpha + \beta)^2 - \alpha\beta = \frac{21}{4}$, determine $k$. [2]

## SECTION C: Short Answer Questions (3 × 3 = 9 Marks)

**Q9.** Find a quadratic polynomial whose zeroes are $2+\sqrt{5}$ and $2-\sqrt{5}$, and verify that the sum and product match the coefficients. [3]

**Q10.** If $\alpha, \beta$ are zeroes of $x^2 - 5x + k$ such that $\alpha - \beta = 1$, find the value of $k$. [3]

**Q11.** If the sum of the zeroes of $f(x) = 2x^2 - 3kx + 5$ is 6, find the value of $k$ and the product of zeroes. [3]

## SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

**Q12.** If the roots of the equation $(x-b)(x-c) + (x-c)(x-a) + (x-a)(x-b) = 0$ are equal real numbers, prove that $a = b = c$. [5]

**Q13.** If the zeroes of the polynomial $x^2 - bx + c$ are two consecutive positive integers, prove that $b^2 - 4c = 1$. Hence find the zeroes if $b = 5$. [5]

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